

Notes: Sweeting, Roberts, and Gedge (JPE, 2020)

A Model of Dynamic Limit Pricing with an Application to the Airline Industry

Contents

1	Big picture	3
2	Data and measurement (key objects)	4
2.1	Empirical background and observed data	4
2.2	Markets, carriers, and phase definitions	4
2.3	Key descriptive objects	5
2.4	Unobservables and timing	5
3	Models and empirical specifications (all equations + interpretation)	6
3.1	Dynamic limit pricing game	6
3.2	Equilibrium theorem and separating pricing	6
3.3	Why dynamics matter: nonmonotonic shading	7
3.4	Reduced-form empirical specification	7
3.5	Demand system and marginal costs	8
3.6	Entry cost calibration and market-size rescaling	9
3.7	Welfare and subsidy counterfactuals	9
4	Identification and instruments (what makes the estimates credible)	10
4.1	What is hard to identify	10
4.2	How the reduced-form evidence identifies deterrence	11
4.3	Why the first-stage probability measure is plausibly exogenous	11
4.4	Why alternative explanations are less plausible	11
4.5	Relation to the literature	12
5	Tables and figures	12
5.1	Table 1 and Table 2: route samples and phase comparisons	12

5.2	Figure 1: dynamic shading logic	13
5.3	Table 3: nonmonotonicity in prices, not capacity	13
5.4	Table 4 and Table 5: demand estimates and marginal cost dynamics	14
5.5	Table 6 and Table 7: rescaled market size and entry cost calibration	14
5.6	Figure 2 and Figure 3: fit of entry and price-decline patterns	15
5.7	Table 8 and Figure 4: welfare and subsidies	15
6	Short summary	16
7	Conclusion	17
7.1	Core conclusion (what the paper establishes)	17
7.2	Economic intuition	17
7.3	Takeaway	17

1 Big picture

- **Question.** Can a dynamic signaling model of limit pricing explain why incumbent airlines cut fares sharply when Southwest becomes a potential entrant, even before Southwest actually enters the route?
- **Setting.** The paper studies airport-pair routes in U.S. domestic aviation, focusing on markets with a dominant incumbent carrier and Southwest as a potential entrant.
- **Core challenge.** The classic Milgrom-Roberts two-period signaling model explains why an incumbent might cut price once to deter entry, but it does not directly explain repeated price cuts over many periods. In practice, Southwest may remain a potential entrant for several quarters or years.
- **Main theoretical result.** The paper builds a dynamic limit-pricing model with serially correlated but not perfectly persistent incumbent marginal costs. Under a D1 refinement, the model has a unique Markov perfect Bayesian equilibrium with fully separating price strategies.
- **Why this matters.** The model makes dynamic asymmetric-information games empirically usable. It shows that signaling can generate large price cuts even when the incumbent's private information has only a modest effect on the entrant's short-run entry probability.
- **Main empirical question.** Are the large "Southwest effect" price cuts better understood as dynamic limit pricing rather than as capacity expansion, declining load factors, airport improvements, or customer-loyalty building?
- **Main reduced-form result.** In dominant-incumbent markets, incumbent price declines are *nonmonotonic* in the probability that Southwest enters. Price cuts are largest in markets with intermediate entry probabilities. That is the qualitative pattern predicted by the model.
- **Main calibration result.** Without using any data on phase-2 price cuts, the calibrated model predicts price declines that are quantitatively close to those observed in the data. The model's largest predicted decline is about **17%**; the largest reduced-form estimate is about **15%**.
- **Main welfare result.** Across the 109 dominant-incumbent markets in the sample, limit pricing increases the present discounted value of consumer surplus by about **\$592 million** and reduces incumbent profits by about **\$86 million**, implying a net welfare gain of a bit more than **\$500 million**.
- **Policy result.** Small entry subsidies to Southwest can raise welfare substantially, especially in low-probability markets, because the subsidy can both increase actual entry and induce incumbents to cut prices more while entry is still only threatened.
- **Economic takeaway.** Dynamic signaling can be very powerful. The incumbent may cut prices a lot not because today's entry probability changes dramatically, but because deterring entry today can preserve many future periods of monopoly.

2 Data and measurement (key objects)

2.1 Empirical background and observed data

- **Industry fact.** Potential competition from Southwest generated the largest documented price effects of potential entry in any industry. Prior work found price reductions of roughly 19%–33% when Southwest served both endpoint airports without yet flying the route.
- **Primary data sources.**
 - The U.S. Department of Transportation’s DB1 Origin-Destination Survey, a quarterly 10% sample of domestic tickets.
 - The DOT’s T100 segment database, which reports monthly flights, capacity, and passengers by carrier and segment.
- **Sample period.** 1993:1 to 2010:4, for a total of 72 quarters.
- **Basic cleaning.** The authors drop very high- and very low-fare itineraries and trips with more than one connection in either direction.
- **Unit of analysis.** A market is a *nondirectional airport pair*. The analysis keeps airport pairs with at least 50 average round-trip passengers per quarter and distance of at least 300 miles.

2.2 Markets, carriers, and phase definitions

- **Southwest actual entry.** Southwest is treated as entering a route once it has at least 65 flights per quarter in T100 and at least 150 direct DB1 passengers on the route.
- **Southwest potential entry.** Southwest is treated as a potential entrant once it serves at least one nonstop route out of *each* endpoint airport.
- **Full sample.** There are **1,542** airport-pair markets in which Southwest becomes a potential entrant after the first quarter of the sample and before 2009:4. Southwest actually enters **337** of them during the sample.
- **Dominant-incumbent sample.** The paper focuses on routes where one carrier is a dominant incumbent before Southwest enters. A carrier must be active in most preentry quarters, hold most direct traffic, and face only negligible actual Southwest route traffic while Southwest is a potential entrant.
- **Dominant-incumbent counts.**
 - **109** markets have a dominant incumbent before Southwest enters.
 - **65** of these have observations both before Southwest becomes a potential entrant and after it becomes a potential entrant but before actual entry. These are the key markets for identifying price responses to threatened entry.
- **Phase definitions.**
 - **Phase 1:** before Southwest is a potential entrant.
 - **Phase 2:** Southwest is a potential entrant but has not yet entered.
 - **Phase 3:** after Southwest has entered.

2.3 Key descriptive objects

- **Route composition (Table 1).** Relative to the full sample, dominant-incumbent markets are shorter, have endpoints more likely to be primary airports in large metropolitan areas, and are larger according to the paper’s gravity-style market-size measure.
- **Selected Table 1 facts.**
 - Full sample mean endpoint population: about **2.51 million**.
 - 109 dominant-incumbent markets mean endpoint population: about **2.83 million**.
 - Full sample mean round-trip distance: about **2,525 miles**.
 - 109 dominant-incumbent markets mean round-trip distance: about **1,258 miles**.
- **Phase-level pricing patterns (Table 2 and text).** In the 65 markets used for price-threat identification, incumbents lower prices by almost **\$90** on average, around **15%**, when Southwest becomes a potential entrant. Actual entry lowers prices by another roughly **30%–45%** relative to far-pre-threat quarters.
- **Traffic composition.** During phase 2, over **80%** of the incumbent’s passengers in these markets are connecting passengers. This is important because the paper argues that connecting traffic can make marginal cost opaque to Southwest.
- **Phase-2 competition from Southwest is limited.** While Southwest serves the endpoint airports in phase 2, it carries only a small share of passengers on the threatened route through connecting service. So the phase-2 effect is mostly about *threat*, not actual route competition.

2.4 Unobservables and timing

- **Incumbent private information.** Before entry, the incumbent’s current marginal cost $c_{I,t}$ is private information.
- **Entrant private information.** The entrant observes a private sunk entry cost draw k_t each period.
- **Timing in each preentry period.**
 1. The incumbent sets a price $p_{I,t}$.
 2. The entrant observes that price and its entry cost draw, then decides whether to enter.
 3. The incumbent’s marginal cost evolves according to a Markov process.
- **Postentry timing.** Once entry occurs, the firms compete under complete information in a static duopoly each period, and the entrant remains active thereafter.
- **Why timing matters.** This timing is what gives price a signaling role. The entrant learns about the incumbent’s toughness from current and past prices before deciding whether to pay the sunk entry cost.

3 Models and empirical specifications (all equations + interpretation)

3.1 Dynamic limit pricing game

- There are two firms: incumbent I and potential entrant E .
- The incumbent is always active.
- The entrant can pay a sunk cost to enter and then becomes active from the next period onward.
- Consumer demand is static within a period.

The incumbent's monopoly profit in a preentry period is

$$\pi_I^M(p_I, c_I) = q^M(p_I)(p_I - c_I),$$

where $q^M(p_I)$ is monopoly demand.

The incumbent's marginal cost evolves according to a first-order Markov process. In the calibrated version this is

$$c_{I,t} = \rho_{AR} c_{I,t-1} + (1 - \rho_{AR}) \frac{\underline{c} + \bar{c}}{2} + \varepsilon_{I,t},$$

with truncated-normal innovations.

Interpretation. The key dynamic state is the incumbent's private marginal cost. It is persistent enough to make current prices informative about future postentry profits, but not perfectly persistent, so the entrant keeps learning over time.

3.2 Equilibrium theorem and separating pricing

A central result of the paper is that, under the D1 refinement, the model has a unique fully separating equilibrium before entry.

The entrant enters if and only if its sunk cost draw is below a threshold:

$$k_t^*(\hat{c}_{I,t}) = \beta \left(E_t[\Phi_{t+1}^E \mid \hat{c}_{I,t}] - E_t[V_{t+1}^E \mid \hat{c}_{I,t}] \right),$$

where:

- $\hat{c}_{I,t}$ is the entrant's posterior belief about the incumbent's current cost,
- V_{t+1}^E is the entrant's continuation value if it stays out,
- Φ_{t+1}^E is the entrant's continuation value if it enters and becomes a duopolist next period.

The incumbent's equilibrium pricing schedule solves the differential equation

$$\frac{\partial p_{I,t}^*}{\partial c_{I,t}} = \frac{\beta g(k_t^*(c_{I,t})) (\partial k_t^*(c_{I,t}) / \partial c_{I,t}) (E_t[V_{t+1}^I \mid c_{I,t}] - E_t[\Phi_{t+1}^I \mid c_{I,t}])}{q^M(p_{I,t}) + (\partial q^M(p_{I,t}) / \partial p_{I,t})(p_{I,t} - c_{I,t})},$$

with boundary condition

$$p_{I,t}^*(\bar{c}_I) = p_I^{\text{static monopoly}}(\bar{c}_I).$$

- Lower-cost incumbents can cut price more cheaply, so they signal toughness by charging lower prices.
- The pricing function is strictly increasing in cost, so prices perfectly reveal type on the equilibrium path.
- Because pricing is fully separating, the entrant’s equilibrium entry decision can be solved using the same entry rule as under complete information.

Interpretation. The denominator is the current profit loss from lowering price. The numerator is the dynamic gain from lowering the probability of entry. Bigger continuation-value gains, more persistent costs, or a more sensitive entry threshold all make limit pricing steeper.

3.3 Why dynamics matter: nonmonotonic shading

The paper rewrites the pricing ODE in a two-period case to show that shading becomes stronger when:

- the incumbent loses a lot from becoming a duopolist,
- the entrant’s entry probability is sensitive to the incumbent’s type,
- or current profit is flat around the static monopoly price, making a price cut relatively cheap.

The calibration assumptions used for the paper’s illustrative figures are:

1. demand is nested logit with market size entering multiplicatively,
2. incumbent marginal cost follows a truncated-normal AR(1),
3. entrant fixed costs are drawn from a truncated normal distribution.

Figure 1 generates several important comparative statics:

- **More persistence means more shading.** With $\rho_{AR} = 0.97$, current price is very informative about future toughness, so the incumbent cuts price more than when $\rho_{AR} = 0.7$.
- **Shading is largest at intermediate entry probabilities.** The entrant’s decision is most sensitive to incumbent type in these markets, so signaling incentives are strongest there.
- **Dynamic continuation values matter a lot.** In low-entry-probability markets, deterring entry today can preserve many future monopoly periods. This makes the dynamic model generate much larger price cuts than a two-period model.

A striking quantitative illustration from Figure 1 is that for market size $M = 20,000$, the incumbent’s cost changes the probability of entry only from **0.143** to **0.119**, yet the average equilibrium price cut is still about **11.6%**. The dynamic gain from preserving future monopoly is what makes such a large cut rational.

3.4 Reduced-form empirical specification

The core reduced-form specification is

$$\text{Price}_{jmt} = \gamma_{jm} + \tau_t + \alpha X_{jmt} + \beta_0 SWPE_{mt} + \beta_1 \hat{r}_m SWPE_{mt} + \beta_2 \hat{r}_m^2 SWPE_{mt} + \varepsilon_{jmt},$$

where:

- $SWPE_{mt}$ is an indicator that Southwest is a potential entrant in market m at time t but has not yet entered,
- $\hat{r}_m$ is the predicted probability that Southwest enters within four quarters of becoming a potential entrant,
- γ_{jm} are market-incumbent fixed effects,
- τ_t are quarter fixed effects,
- and X_{jmt} includes time-varying controls such as competitors and jet-fuel-price interactions.

Interpretation. The phase-2 price effect is allowed to vary flexibly with the route's ex ante entry probability. The model predicts $\beta_1 < 0$ and $\beta_2 > 0$: the phase-2 price cut should be strongest at intermediate entry probabilities.

The same specification is also estimated using:

- log capacity,
- log passengers,
- and log load factor

as dependent variables, to test whether incumbents respond to threatened entry through capacity or other channels instead of price.

3.5 Demand system and marginal costs

The paper estimates a one-level nested logit demand system in which the incumbent and Southwest are in the same nest once entry occurs. Indirect utility has the form

$$u_{i,j,m,t} = v_{j,m,t} - \alpha p_{j,m,t} + \sigma \text{FLY}_{i,m,t} + (1 - \lambda) \varepsilon_{i,j,m,t},$$

where $v_{j,m,t}$ contains carrier fixed effects, route characteristics, seasonal terms, and measures of competing service.

- **Demand estimates (2SLS).**
 - Price coefficient $\hat{\alpha} \approx 2.446$.
 - Nesting parameter $\hat{\lambda} \approx 0.793$.
- These estimates imply that the incumbent and Southwest are close substitutes after entry.
- The average incumbent own-price elasticity in phase 3 is about -2.92 .

Marginal costs are backed out from static first-order conditions in phases 1 and 3, under the assumption that limit pricing occurs only in phase 2. The estimated cost process is

$$mc_{j,t} = \rho_{AR} mc_{j,t-1} + X_{j,t} \gamma + \mu_t + \mu_j + \varepsilon_{j,t}.$$

- The average implied incumbent marginal cost is about **\$258**.

- Southwest’s implied marginal cost is about **31% lower** than the incumbent’s in phase 3, roughly **5.4 cents per mile** lower.
- The pooled 2SLS estimate of serial correlation is about $\rho_{AR} = 0.974$.

Interpretation. Southwest is modeled as the low-cost entrant. Persistent incumbent marginal cost is the private information that gets signaled through price.

3.6 Entry cost calibration and market-size rescaling

The authors estimate a Weibull hazard model for Southwest’s route entry decisions once Southwest becomes a potential entrant. They then rescale market size so that the entry hazard can be summarized by a single transformed market-size variable.

- This rescaling absorbs route features such as airport importance, concentration, and network fit.
- Example: Las Vegas–Omaha has a smaller raw market size than Omaha–St. Louis, but a *higher* rescaled market size because Las Vegas is a leisure destination and Southwest focus city.

Entry costs are then assumed to be normal with:

- a mean that depends on rescaled market size,
- a common standard deviation,
- and a small upward drift over time to match declining empirical entry hazards.

Key calibration facts:

- For the median market, mean initial entry cost is about **\$44.2 million**.
- The estimated standard deviation of entry costs is about **\$204,000**.
- Entry costs rise only slightly over time, but that is enough to generate a large decline in hazard rates because entry decisions are very sensitive near the margin.

Interpretation. The entry cost is best thought of as including discounted future fixed costs, such as the fixed costs of capacity and airport presence, not just a one-time setup charge.

3.7 Welfare and subsidy counterfactuals

The calibrated model is used for two main counterfactuals.

1. Asymmetric information versus complete information

The paper compares equilibrium outcomes under:

- **Asymmetric information:** incumbent marginal cost is private and signaling occurs.
- **Complete information:** Southwest knows incumbent costs, so there is no need for signaling and no limit pricing.

Across the 109 dominant-incumbent markets, limit pricing implies:

- present discounted value of reduced prices: about **\$538.7 million**,
- gain in consumer surplus: about **\$592.3 million**,
- loss in incumbent profits: about **\$86.2 million**.

Interpretation. Limit pricing is not just a transfer from firms to consumers. It also increases total welfare because price cuts occur while the route is still a monopoly market and move fares closer to competitive levels.

2. Small entry subsidies

The policy experiment gives Southwest a fixed subsidy whenever it provides nonstop service on a route.

- Even a small subsidy can materially increase welfare.
- In low-probability markets, the subsidy may raise consumer surplus by more under asymmetric information than under complete information, because it also intensifies incumbent limit pricing before entry occurs.

For Hartford–Minneapolis, a low-probability market, the paper finds:

- baseline first-period shading around **\$20** or **3.8%**,
- with a \$1,000-per-quarter subsidy, first-period shading rises to about **\$27** or **5.3%**,
- the expected cost to the government is only about **\$7,000**,
- and the present value of consumer-surplus gains under asymmetric information is about **\$8.55 million**.

Interpretation. Targeted subsidies can create large benefits even when actual entry remains unlikely, because a small increase in entry probability can induce the incumbent to cut prices materially.

4 Identification and instruments (what makes the estimates credible)

4.1 What is hard to identify

- A price cut after Southwest becomes a potential entrant does not automatically imply signaling or deterrence.
- It could reflect airport improvements, capacity expansion, lower load factors, stronger connecting competition, or loyalty building.
- The empirical challenge is to isolate the *threat-of-entry* channel from these alternative mechanisms.

4.2 How the reduced-form evidence identifies deterrence

- The dominant-incumbent markets are chosen because they come closest to the one-incumbent/one-entrant structure assumed in the theoretical model.
- The paper does not just ask whether prices fall in phase 2. That fact was already known.
- Instead, it asks whether the *pattern* of price cuts across markets matches the model’s prediction.

The distinctive prediction is nonmonotonicity:

- if signaling incentives are strongest when entry probability is intermediate, price cuts should be largest there;
- this pattern is harder to explain with stories based purely on changing marginal cost or actual competition.

4.3 Why the first-stage probability measure is plausibly exogenous

- The first-stage entry-probability model is estimated on the **full sample excluding dominant-incumbent markets**. So the same markets are not used both to estimate threat probabilities and to estimate price responses.
- Variables based on passenger flows are measured using **phase-1 quarters more than one year before phase 2**, reducing the risk that phase-2 price cuts contaminate the first-stage covariates.
- The first-stage includes route size, concentration, network fit, and time controls, all of which are natural determinants of Southwest entry.

Interpretation. The authors try to construct $\hat{r}_m$ from variables that shift the attractiveness of entry but should not themselves be consequences of phase-2 strategic pricing.

4.4 Why alternative explanations are less plausible

The paper directly evaluates several alternatives.

- **Airport improvements story.** If airport developments both lower incumbent costs and attract Southwest, the nonmonotonic price pattern could be spurious. The authors show the pattern is robust to airport-by-phase-2 fixed effects and find no similar nonmonotonicity at the end of phase 1.
- **Capacity expansion story.** If incumbents invest in capacity to deter Southwest, prices might fall because marginal cost falls. But Table 3 shows no significant phase-2 capacity response of the right form.
- **Load factor story.** If incumbents lose demand, load factors would fall and marginal costs might decline. Instead, traffic and load factors *increase* in intermediate-probability markets, which moves against that explanation.
- **Connecting-service competition.** Southwest carries only a small share of route passengers through connecting service in phase 2, so actual competition is limited before entry.

- **Frequent-flyer loyalty building.** The data show similar nonmonotonic price effects across the price distribution, not just for the most expensive business-travel tickets, and the authors find no evidence that temporary price cuts increase future demand as a loyalty story would predict.

4.5 Relation to the literature

- The paper extends the classic Milgrom-Roberts limit-pricing model to a dynamic setting with repeated signaling.
- It connects to the airline “Southwest effect” literature by offering a structural explanation for preentry price cuts.
- It also contributes to the literature on empirical dynamic games with asymmetric information by showing that a useful and tractable MPBE can be characterized and calibrated.

5 Tables and figures

5.1 Table 1 and Table 2: route samples and phase comparisons

TABLE 1
COMPARISON OF MARKETS IN THE FULL SAMPLE AND DOMINANT INCUMBENT SAMPLES

	DOMINANT INCUMBENT SAMPLES					
	FULL SAMPLE		109 Markets		65 Markets	
	Mean	Standard Deviation	Mean	Standard Deviation	Mean	Standard Deviation
Mean end point population (millions)	2.509	1.918	2.834	1.923	3.218	2.112
Round-trip distance (miles)	2,525.11	1,352.57	1,257.57	743.08	1,344.5	798.75
Constructed market size measure	33,005	46,389	65,637	68,589	52,325	62,642
Origin or destination is a:						
Primary airport in multiairport MSA	.186	.389	.321	.469	.262	.443
Secondary airport in multiairport MSA	.316	.465	.321	.469	.369	.486
Airport in big city	.643	.479	.844	.364	.877	.331
Leisure destination	.108	.311	.110	.314	.092	.292
Slot-controlled airport	.039	.193	.064	.246	.108	.312
Number of markets	1,542		109		65	

NOTE.—We define leisure destinations (primarily cities in Florida; Las Vegas; Charleston, SC; and New Orleans) and big cities (top 30 metropolitan statistical areas [MSAs] excluding leisure destinations) following Gerardi and Shapiro (2009). We define New York’s John F. Kennedy, LaGuardia, and Newark; Washington’s Reagan; and Chicago’s O’Hare airports as slot controlled, although slot controls are no longer in place at O’Hare. We identify metropolitan areas with more than one major airport using Wikipedia (http://en.wikipedia.org/wiki/List_of_cities_with_more_than_one_airport) and identify the primary airport as the one with the most passenger traffic in 2012.

TABLE 2
SUMMARY STATISTICS: DOMINANT INCUMBENT SAMPLE

VARIABLE	PHASE 1: $t < t_0$							
	All Markets		Markets with Phase 1 and Phase 2 Observation		Phase 2: $t_0 \leq t < t_1$		Phase 3: $t \geq t_1$	
	Mean	Standard Deviation	Mean	Standard Deviation	Mean	Standard Deviation	Mean	Standard Deviation
Incumbent pricing:								
Yield (average fare per mile)	.516	.331	.527	.339	.452	.323	.311	.166
Average fare	476.77	135.18	514.55	144.44	426.06	117.68	261.83	60.65
Southwest pricing:								
Yield					.296	.150	.235	.072
Average fare					388.50	120.45	214.37	62.63
Passenger shares:								
Incumbent					.840	.122	.461	.198
Southwest	.797	.213	.743	.235	.018	.032	.479	.217
Incumbent capacity and traffic:								
Capacity (seats performed)	75,760	52,459	72,785	49,012	69,770	47,066	90,877	52,314
Segment passengers (including connecting passengers)	46,072	32,141	44,174	29,814	48,478	31,677	64,585	38,585
Load factor	.612	.104	.618	.106	.710	.121	.705	.081
Proportion of passengers connecting	.837	.114	.847	.109	.830	.115	.774	.107
Southwest capacity and traffic:								
Capacity (seats performed)							80,751	62,207
Segment passengers (including connecting passengers)							52,713	39,195
Load factor							.651	.083
Proportion of passengers connecting							.701	.101
Number of markets	109		65		65		59	

Read:

- Table 1 shows that the dominant-incumbent markets are not a random set of routes. They are shorter, more urban, and strategically important. That is exactly where one would expect Southwest threats to matter most.
- The fact that there are only 65 routes with both clean pre-threat and threat-only periods is important. The main identification comes from a relatively focused set of markets, not from the entire universe of routes.
- Table 2 is the key descriptive table for the paper’s empirical motivation: incumbent prices fall substantially already in phase 2, before Southwest actually enters.

- Table 2 also shows that incumbents carry many connecting passengers and that Southwest has only limited route presence in phase 2. This supports the asymmetric-information and threat-of-entry interpretation.

5.2 Figure 1: dynamic shading logic

Read:

- Panels A and B show how price schedules lie below static monopoly prices: this is the model’s notion of limit pricing or “shading.”
- Panels C and D contain the paper’s central comparative-static intuition: shading is largest at intermediate entry probabilities, not where entry is almost impossible or almost certain.
- Panels E and F explain why the dynamic model is quantitatively powerful. When future entry is still possible for many periods, even modest current changes in entry probability can justify sizable current price cuts.
- This figure is the bridge between the theory and the reduced-form evidence in section IV.

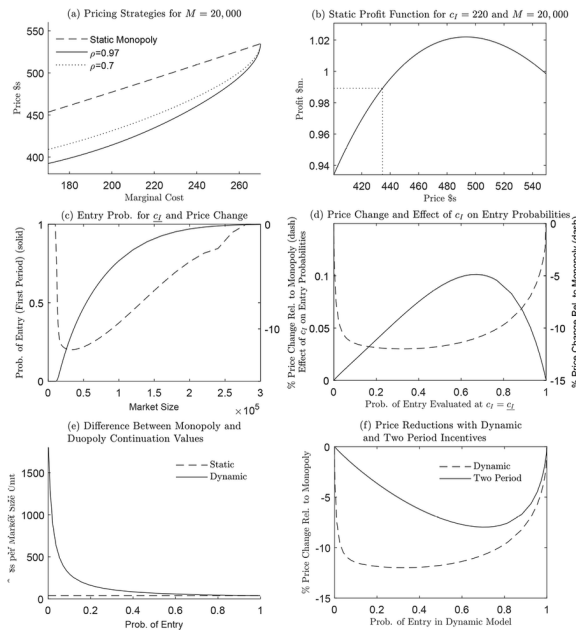


FIG. 1.—Relationship between market size, entry probabilities, and shading in the dynamic limit pricing model.

TABLE 3
SECOND-STAGE ESTIMATES OF THE RELATIONSHIP BETWEEN THE PROBABILITY
THAT SOUTHWEST ENTERS AND CHANGES IN INCUMBENT PRICES,
CAPACITIES, SEGMENT TRAFFIC, AND LOAD FACTORS

Dependent Variable	Log Price (1)	Yield (2)	Log Capacity (3)	Log Passengers (4)	Log Load Factor (5)
$SWPE_{n,t}$	-.043* (.023)	-.002 (.014)	.068 (.043)	.144*** (.044)	.076*** (.017)
$\hat{\rho}_n \cdot SWPE_{n,t}$	-.693*** (.182)	-.732*** (.142)	.040 (.362)	.578 (.413)	.538*** (.142)
$\hat{\rho}_n^2 \cdot SWPE_{n,t}$	1.169*** (.256)	1.046*** (.219)	-.820 (.619)	-2.053*** (.749)	-1.233*** (.236)
Observations	3,884	3,884	3,400	3,400	3,400

NOTE.—Heteroskedasticity-robust Newey-West standard errors allowing for one-period serial correlation and corrected for first-stage approximation error in the entry probabilities are given in parentheses. The number of observations reflect differences in the coverage and reporting in the DB1 and T100 data during our sample period. The predicted probability that Southwest enters within four quarters of becoming a potential entrant is based on estimates reported in app. D.1, and these specifications include market-incumbent and quarter fixed effects and time-varying controls listed in the main text.

* Significant at the 10% level.
*** Significant at the 1% level.

5.3 Table 3: nonmonotonicity in prices, not capacity

Read:

- In the log-price regression, the phase-2 effect is negative, the interaction with entry probability is strongly negative, and the squared interaction is positive. This is the U-shaped relationship the model predicts.
- The largest predicted price decline is just under **15%**, at intermediate entry probabilities.

- Capacity does not move in the way a capacity-deterrence story would require.
- By contrast, total passengers and load factors increase in intermediate-probability markets. This makes it hard to tell a story in which prices fall because the incumbent’s marginal cost is falling.

5.4 Table 4 and Table 5: demand estimates and marginal cost dynamics

TABLE 4
NESTED LOGIT DEMAND: PRICE AND NESTING PARAMETERS

	OLS	2SLS
Fare (\$, hundreds; $\hat{\alpha}$)	-.317*** (.011)	-.446*** (.034)
Inside share ($\hat{\lambda}$)	.748*** (.033)	.793*** (.072)
Observations	6,037	6,037
R^2	.301	

NOTE.—Specification also includes a linear time trend, quarter-of-year dummies, dummies for Southwest and the major incumbent carriers, market characteristics (distance, distance², indicators for whether the route includes a carrier’s hub or a leisure destination or is in a city with another major airport), and dummies for the number of competitors offering direct service. The instruments used for 2SLS are described in the main text. Robust standard errors are given in parentheses. OLS = ordinary least squares.

*** Significant at the 1% level.

TABLE 5
MARGINAL COST EVOLUTION: ESTIMATES OF SERIAL CORRELATION

	OLS All Carriers (1)	2SLS All Carriers (2)	2SLS Southwest (3)	2SLS Incumbents (4)
MC per mile, $\widehat{mc}_{m,t-1}$	.916*** (.037)	.974*** (.013)	.978*** (.039)	.962*** (.012)
Observations	5,658	4,710	1,492	3,218
R^2	.834			

NOTE.—The dependent variable is MC per mile, $\widehat{mc}_{m,t}$, carrier j ’s computed marginal cost (MC) per mile in market m in quarter t . The specification also includes market characteristics (market size, average population, distance, and a dummy for whether one of the airports is slot constrained), quarter dummies, carrier dummies, and the lagged price of jet fuel interacted with route distance. In columns 2–4, we use the third through fifth lags of marginal cost per mile to instrument for lagged marginal costs. Robust standard errors, corrected for the uncertainty in the demand estimates, are given in parentheses. OLS = ordinary least squares.

*** Significant at the 1% level.

Read:

- Table 4 shows the paper’s preferred 2SLS nested-logit demand estimates. The nesting parameter is high, which means Southwest and the incumbent are close substitutes once Southwest enters.
- The implied own-price elasticity of about -2.92 is economically plausible and large enough that postentry competition matters a lot.
- Table 5 shows strong serial correlation in marginal cost, with ρ around **0.97**. This is crucial because the whole signaling mechanism relies on today’s price being informative about future postentry profitability.
- Southwest’s estimated marginal cost is substantially lower than that of legacy carriers, which fits the economic narrative behind the Southwest effect.

5.5 Table 6 and Table 7: rescaled market size and entry cost calibration

TABLE 6
WEIBULL HAZARD MODEL: PREDICTED HAZARD RATES AND RESCALED MARKET SIZE FOR THREE MARKETS

Route	Original Market Size	$\widehat{h}_{m,2}$	$\widehat{h}_{m,10}$	Rescaled Market Size
Las Vegas–Omaha	19,820	.208	.098	46,462
Omaha–St. Louis	36,568	.092	.042	37,402
Minneapolis–Omaha	38,763	.037	.017	27,948

NOTE.—The table shows, for three example markets, the original market size (constructed as described in app. C.2); estimated probabilities of entry after two and 10 quarters (“hazard rates”), conditional on not having entered in an earlier period; and our rescaled market size that captures all of the observed variables entering the linear index of the hazard model. The hazard entry probabilities are calculated by finding the survival probability, $S_{m,t}$, for each quarter and then calculating entry probability as $(S_{m,t-1} - S_{m,t})/S_{m,t-1}$.

TABLE 7
CALIBRATION OF ENTRY COSTS: PARAMETER ESTIMATES

	Constant	Log (Rescaled Market Size) (10,000)	Rescaled Market Size (10,000)	(Rescaled Market Size) (10,000)
Log ($\frac{\text{Mean Entry Costs}}{10 \text{ million}}$)	2.524 (.016)	.968 (.017)		
Log (Standard Deviation of Entry Costs) (1 million)	-1.588 (.511)			
Log(γ)	-6.790 (1.745)		9.050 (7.174)	-12.819 (6.135)
γ	-1.940 (1.368)		-4.545 (4.688)	10.626 (4.014)

NOTE.—Parameters minimize the sum of squared residuals comparing predicted hazard probabilities of entry with ones estimated from a Weibull hazard model based on 21 dominant carrier markets. The rescaling of market size is described in the main text. Standard errors in parentheses reflect uncertainty only from the calibration stage of estimation, not from estimation of the Weibull hazard model, the carrier demand, or marginal cost models or from the rescaling of market size.

Read:

- Table 6 explains how the rescaled market-size variable works. It is not raw demand alone that matters for entry but also how the route fits Southwest’s network and airport characteristics.
- Table 7 is the entry-calibration table. The paper does not estimate the entry process by matching price declines; it estimates the entry-cost distribution by matching entry hazards.
- This separation is a credibility strength: price cuts are a test of the calibrated model, not an input used to force the model to fit.

5.6 Figure 2 and Figure 3: fit of entry and price-decline patterns

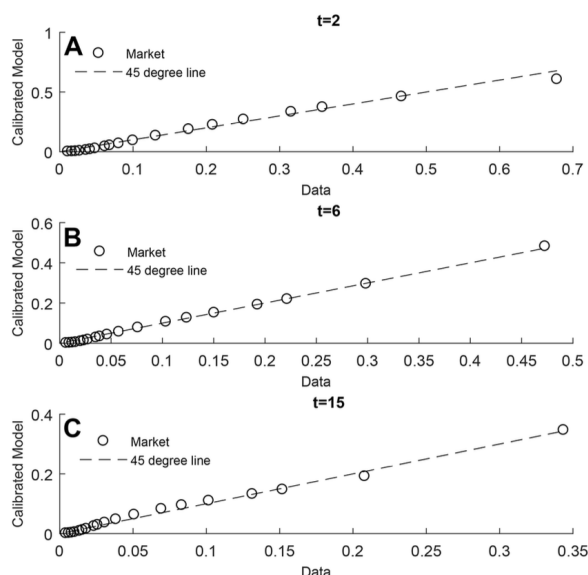


FIG. 2.—Match of empirical entry probabilities (conditional on entry having not already occurred) and the probabilities predicted by the calibrated model for the 21 markets used in the calibration.

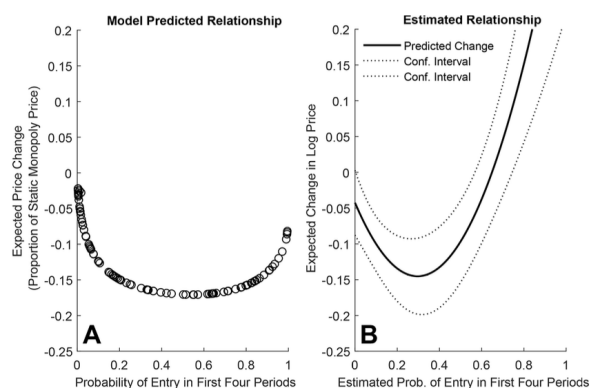


FIG. 3.—Predicted and estimated relationships between price changes and the probability of entry.

Read:

- Figure 2 shows the calibrated model matches empirical entry probabilities across markets and horizons ($t = 2, 6, 15$) quite closely.
- Figure 3 is the main model-validation figure. Figure 3A shows the model-implied relationship between price cuts and entry probabilities; Figure 3B shows the reduced-form relationship from the data.
- The qualitative match is very strong: both display clear nonmonotonicity, with the biggest price declines in intermediate-probability markets.
- The model’s maximum predicted decline is about **17%**, while the data’s peak decline is about **15%**. That is a strong quantitative success given that phase-2 price changes were not used in the calibration.

5.7 Table 8 and Figure 4: welfare and subsidies

Read:

	Market			
	Hartford– Minneapolis	Manchester, New Hampshire– Philadelphia	Las Vegas– San Jose	All Dominant Incumbent Markets
Market rank (by rescaled market size, out of 109 markets)	20	65	105	
Actual market size	33,828	45,481	136,861	Total: 5.7 million
Model predictions:				
Second quarter:				
Probability of entry (if no entry in first quarter)	.002	.095	.606	Mean: .151
Shading (\$, %; relative to static monopoly price)	20.07, 3.8	86.49, 16.6	57.04, 10.9	Mean: 59.85, 11.4
Twentieth quarter:				
Probability of entry (if no entry previously)	.002	.046	.266	Mean: .050
Shading (\$, %; relative to static monopoly price)	16.87, 3.2	69.23, 13.3	117.06, 22.4	Mean: 52.44, 10.1
Probability of entry within four quarters	.010	.339	.973	Mean: .355
Welfare effects of limit pricing (relative to complete information):				
PDV of reduced prices (\$, millions)	4.32	6.20	.76	Total: \$38.74
PDV of change in consumer surplus (\$, millions)	4.38	7.07	.84	Total: \$62.27
PDV of change in incumbent profits (\$, millions)	−.18	−1.35	−.13	Total: −\$6.18
Impact of a \$1,000 per quarter subsidy:				
Probability of entry within four quarters	.018	.372	.974	Mean: .374
Second-period shading (\$, %)	27.40, 5.3	89.62, 16.3	60.98, 10.9	Mean: 54.50, 10.5
PDV of the cost to government (\$, millions)	.007	.039	.046	Total: \$3.97
Changes under complete information:				
PDV of change in consumer surplus (\$, millions)	7.196	4.127	.078	Total: \$77.74
PDV of change in incumbent profits (\$, millions)	−3.880	−2.239	−.042	Total: −\$14.15
Changes under asymmetric information:				
PDV of consumer surplus gain (\$, millions)	8.553	3.264	.059	Total: \$53.91
PDV of change in incumbent profits (\$, millions)	−4.038	−2.083	−.039	Total: −\$12.14

NOTE.—PDV = present discounted value.

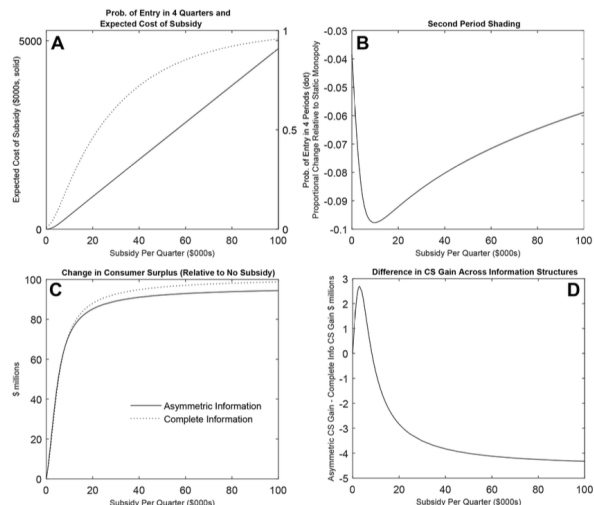


FIG. 4.—Predicted effect of fixed-cost subsidies for Southwest on entry, incumbent shading, and consumer surplus in the Hartford–Minneapolis market. CS = consumer surplus.

- Table 8 translates the pricing results into welfare. Even modest phase-2 price shading can create large present discounted consumer gains because the low prices persist while entry remains only a threat.
- The aggregate consumer-surplus gain of about **\$592 million** from just 109 routes shows that threatened entry can matter a lot even when actual entry is rare.
- Figure 4 is especially important for policy. In a low-probability market, raising entry probability a little can produce a lot of consumer surplus because it both increases actual entry and induces more incumbent shading.
- The figure also shows a key nonlinearity: at very high subsidy levels, entry becomes almost certain, shading disappears quickly, and the extra value of asymmetric-information effects falls.

6 Short summary

- **Method.** The paper develops a dynamic signaling model of limit pricing with a unique MPBE and calibrates it to airline data without using the key preentry price-cut moments as calibration targets.
- **Identification.** The empirical signature is a nonmonotonic relationship between price cuts and Southwest entry probabilities in dominant-incumbent markets, together with evidence against capacity, cost, and loyalty alternatives.
- **Application.** In U.S. airline markets, dominant incumbents cut prices substantially when Southwest becomes a potential entrant, even before actual entry.
- **Result.** The calibrated model matches this pattern well, implies sizable welfare gains for consumers from limit pricing, and predicts that even small entry subsidies can have large consumer benefits in low-probability markets.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- A dynamic limit-pricing model with persistent asymmetric information can be both theoretically sharp and empirically useful.
- The paper shows that the Southwest effect is plausibly a signaling-based entry-deterrence phenomenon, not merely a static response to actual competition.
- The dynamic extension is essential because it explains why prices stay low when entry remains only a threat for a long time.

7.2 Economic intuition

- A low-cost incumbent can afford to cut price more than a high-cost incumbent, so price becomes a credible signal of toughness.
- The entrant is most responsive to the signal when entry is neither nearly certain nor nearly impossible, which is why the biggest price cuts arise in intermediate-probability markets.
- Dynamics amplify the signal because deterring entry today protects many future periods of monopoly. That makes a small change in today's entry hazard potentially very valuable.

7.3 Takeaway

This paper is important both methodologically and substantively. Methodologically, it shows that a dynamic asymmetric-information model can generate a unique, tractable equilibrium and can be disciplined with real data. Substantively, it provides a persuasive explanation for why dominant airlines cut fares before Southwest actually enters. The bigger lesson is that the threat of entry can reshape prices long before market structure changes, and dynamic signaling can be a quantitatively important reason why.